Why not use the divergence matrix for spectral scale selection?

A corrected endpoint argument and its Gaussian-KDE specialization

September 10, 2026

Abstract

This note makes precise the endpoint bias that arises when the Perron vector is computed from a dissimilarity matrix rather than from a similarity matrix. The ambient convexity assumptions in the original proposed lemma are insufficient: the observations must also be ordered along a common geodesic. After proving the corrected geodesic statement, we give a result tailored to Gaussian kernel density estimators. For increasing bandwidths, the squared L^2 divergence matrix is a strict Robinson dissimilarity, and its Perron vector attains its maximum only at one or both extreme bandwidths. Thus the divergence matrix systematically rewards extremeness in this setting, whereas a decreasing similarity transform reverses the direction of comparison.

1 What is missing from the original lemma?

The conclusion cannot follow merely from convexity of the ambient set. The indices 1 and N have no geometric meaning unless the sample is assumed to lie, in that order, on a single curve joining x_1 to x_N.

Remark 1.1 (A counterexample to the original assumptions). Let $M = \mathbb{R}^2$ with the Euclidean distance and let x_1, x_2, x_3 be the vertices of an equilateral triangle. Distance to a fixed point is convex along every Euclidean geodesic, and the convex hull of the three points is convex. However, the distance matrix is

$$D = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix},$$

whose Perron vector is proportional to $(1, 1, 1)^{\mathsf{T}}$. Hence $\arg\max_i v_i = \{1, 2, 3\}$, not a subset of $\{1, 3\}$. The missing assumption is that all sampled points lie on a common geodesic, with x_1 and x_N as its sampled endpoints.

The following version also applies to a possibly asymmetric divergence. For such a matrix, "Perron vector" means the positive right Perron vector.

Lemma 1.2 (Ordered-geodesic endpoint lemma). *Let $\gamma : [0, 1] \to M$ be a nonconstant geodesic and let*

$$0 = t_1 < t_2 < \cdots < t_N = 1, \qquad x_i = \gamma(t_i).$$

Let $\Delta : M \times M \to [0, \infty)$ satisfy $\Delta(x, y) = 0$ if and only if $x = y$. Assume that, for every $y \in M$, the map

$$t \longmapsto \Delta(\gamma(t), y)$$

is convex. Set $D_{ij} = \Delta(x_i, x_j)$, and let $v > 0$ satisfy $Dv = \rho v$ for the Perron root ρ. Then

$$\arg\max_{1 \le i \le N} v_i \subseteq \{1, N\}.$$

More precisely, every interior index i satisfies $v_i < \max\{v_1, v_N\}$.

Proof. Because the points are distinct, every off-diagonal entry of D is positive. Thus D is irreducible, $\rho > 0$, and Perron–Frobenius gives a vector $v > 0$. Define

$$F(x) = \sum_{j=1}^{N} \Delta(x, x_j) v_j.$$

The restriction $F \circ \gamma$ is convex. At every sampled point,

$$F(x_i) = \sum_{j=1}^{N} D_{ij} v_j = (Dv)_i = \rho v_i. \tag{1}$$

For an interior index i, convexity of each summand gives

$$F(x_i) \le (1 - t_i) F(x_1) + t_i F(x_N).$$

This inequality is strict. Indeed, the summand with $j = i$ has value zero at x_i, while its endpoint chord has the strictly positive value

$$(1 - t_i)\Delta(x_1, x_i) v_i + t_i \Delta(x_N, x_i) v_i > 0.$$

Consequently,

$$\rho v_i = F(x_i) < (1 - t_i) F(x_1) + t_i F(x_N) \le \max\{F(x_1), F(x_N)\}.$$

Using (1) and dividing by $\rho > 0$ yields $v_i < \max\{v_1, v_N\}$. □

Remark 1.3. For a metric d, the hypotheses of Lemma 1.2 hold whenever distance to every fixed point is geodesically convex along γ. The essential addition to the original statement is the ordered representation $x_i = \gamma(t_i)$. Convexity of a larger set containing unrelated points does not provide the endpoint comparison used in the proof.

2 An acute-chain theorem for squared Hilbert distances

The Gaussian-KDE path is generally not a straight L^2 geodesic, so Lemma 1.2 cannot simply be invoked for that path. A different geometric property supplies the desired conclusion.

Definition 2.1 (Strictly acute chain). *Let H be a real Hilbert space and let $x_1, \dots, x_N \in H$ be distinct. Put $\delta_k = x_{k+1} - x_k$ for $k = 1, \dots, N-1$. The ordered family is called a strictly acute chain if*

$$\langle \delta_k, \delta_\ell \rangle > 0 \qquad \text{for all } k, \ell \in \{1, \dots, N-1\}.$$

Theorem 2.2 (Perron endpoint theorem for an acute chain). *Let $x_1, \dots, x_N$ be a strictly acute chain in a real Hilbert space and set*

$$D_{ij} = \frac{1}{2} \|x_i - x_j\|^2 .$$

If $v > 0$ is the Perron vector of D, then

$$\arg\max_{1 \le i \le N} v_i \subseteq \{1, N\}.$$

Proof. Write $V = \sum_j v_j$, $w_j = v_j/V$, and

$$\mu = \sum_{j=1}^{N} w_j x_j, \qquad \sigma^2 = \sum_{j=1}^{N} w_j \left\|x_j - \mu\right\|^2.$$

The variance decomposition gives, for every i,

$$\begin{aligned}(Dv)_i &= \frac{1}{2}\sum_{j=1}^{N} v_j \left\|x_i - x_j\right\|^2 \\ &= \frac{V}{2}(\left\|x_i - \mu\right\|^2 + \sigma^2). \end{aligned} \tag{2}$$

Since $(Dv)_i = \rho v_i$ and $\rho > 0$, equation (2) shows that the ordering of the entries v_i is exactly the ordering of the squared distances $\left\|x_i - \mu\right\|^2$.

It remains to show that no interior x_i is farthest from μ. The barycenter has the increment representation

$$\mu = x_1 + \sum_{k=1}^{N-1} a_k \delta_k, \qquad a_k = \sum_{j=k+1}^{N} w_j.$$

Strict positivity of the w_j implies

$$1 > a_1 > a_2 > \cdots > a_{N-1} > 0. \tag{3}$$

For $2 \le i \le N-1$, set

$$P_i = \sum_{k<i}(1 - a_k)\delta_k, \qquad Q_i = \sum_{k\ge i} a_k \delta_k.$$

Then $x_i - \mu = P_i - Q_i$. Both vectors are nonzero, all their coefficients are positive, and the acute-chain assumption gives $\langle P_i, Q_i\rangle > 0$. Therefore

$$\left\|x_i - \mu\right\|^2 = \left\|P_i - Q_i\right\|^2 < \left\|P_i + Q_i\right\|^2. \tag{4}$$

The Gram matrix $(\langle \delta_k, \delta_\ell\rangle)_{k,\ell}$ has nonnegative entries. Hence the norm of a nonnegative linear combination of the δ_k is monotone in each coefficient. If $a_{i-1} \ge \frac{1}{2}$, then, by (3), $1 - a_k \le a_k$ for every $k < i$. Thus

$$\left\|P_i + Q_i\right\| \le \left\|\sum_{k=1}^{N-1} a_k \delta_k\right\| = \left\|\mu - x_1\right\|.$$

If $a_{i-1} < \frac{1}{2}$, then $a_k < \frac{1}{2}$ for every $k \ge i$, and hence $a_k \le 1 - a_k$ there. Consequently,

$$\left\|P_i + Q_i\right\| \le \left\|\sum_{k=1}^{N-1} (1 - a_k)\delta_k\right\| = \left\|x_N - \mu\right\|.$$

Combining either inequality with the strict inequality (4) proves

$$\left\|x_i - \mu\right\| < \max\{\left\|x_1 - \mu\right\|, \left\|x_N - \mu\right\|\}.$$

The ordering equivalence from (2) completes the proof. □

3 Gaussian KDEs with squared L^2 divergence

Definition 3.1 (Robinson dissimilarity [1]). *A symmetric dissimilarity matrix D is Robinson in the displayed order if, whenever $i < j < k$,*

$$D_{ik} \geq \max\{D_{ij}, D_{jk}\}.$$

It is strict Robinson *if the inequality is strict for every such triple. Equivalently, entries do not decrease when one moves away from the diagonal in any row or column.*

Let ν be a probability measure on $\mathbb{R}^d$; in the empirical KDE setting, $\nu = n^{-1}\sum_{r=1}^{n}\delta_{X_r}$. For $h > 0$, let

$$p_h = \nu * \varphi_h, \qquad \varphi_h(x) = (2\pi h^2)^{-d/2}\exp\left(-\frac{\|x\|^2}{2h^2}\right).$$

Consider the squared L^2 divergence

$$\mathcal{D}_2(p,q) = \frac{1}{2}\|p-q\|^2_{L^2(\mathbb{R}^d)}.$$

Proposition 3.2 (Gaussian smoothing produces a strictly acute chain). *For $0 < h_1 < \cdots < h_N$, the ordered family $p_{h_1}, \ldots, p_{h_N}$ is a strictly acute chain in $L^2(\mathbb{R}^d)$.*

Proof. Use the Fourier convention $\widehat{f}(\omega) = \int_{\mathbb{R}^d} f(x)\mathrm{e}^{-\mathrm{i}x\cdot\omega}\,dx$. Then

$$\widehat{p}_h(\omega) = \widehat{\nu}(\omega)\exp\left(-\frac{h^2\|\omega\|^2}{2}\right).$$

For $\delta_k = p_{h_{k+1}} - p_{h_k}$, define

$$b_k(\omega) = \exp\left(-\frac{h_{k+1}^2\|\omega\|^2}{2}\right) - \exp\left(-\frac{h_k^2\|\omega\|^2}{2}\right).$$

For $\omega \neq 0$, every $b_k(\omega)$ is strictly negative. Plancherel's identity therefore yields

$$\langle\delta_k, \delta_\ell\rangle_{L^2} = \frac{1}{(2\pi)^d}\int_{\mathbb{R}^d}|\widehat{\nu}(\omega)|^2 b_k(\omega)b_\ell(\omega)\,d\omega > 0. \tag{5}$$

To justify strictness, note that $\widehat{\nu}$ is continuous and $\widehat{\nu}(0) = 1$, so it is nonzero on a neighborhood of the origin. On that neighborhood, away from the single point $\omega = 0$, the integrand in (5) is strictly positive on a set of positive Lebesgue measure. □

Corollary 3.3 (Robinson property and endpoint bias). *Let $0 < h_1 < \cdots < h_N$ and define*

$$D_{ij} = \mathcal{D}_2(p_{h_i}, p_{h_j}) = \frac{1}{2}\left\|p_{h_i} - p_{h_j}\right\|_2^2.$$

Then:

(i) *D is a strict Robinson dissimilarity in the natural bandwidth order: for every $i < j < k$,*

$$D_{ik} > \max\{D_{ij}, D_{jk}\};$$

(ii) *if v is the Perron vector of D, then*

$$\arg\max_{1\leq i\leq N} v_i \subseteq \{1, N\}.$$

Proof. For $i < j < k$, write

$$p_{h_k} - p_{h_i} = A + B, \quad A = p_{h_j} - p_{h_i}, \quad B = p_{h_k} - p_{h_j}.$$

Proposition 3.2 gives $\langle A, B\rangle > 0$, since each is a positive sum of consecutive increments. Hence

$$\|A + B\|^2 = \|A\|^2 + \|B\|^2 + 2\langle A, B\rangle > \max\{\|A\|^2, \|B\|^2\},$$

which proves (i). Part (ii) follows directly from Theorem 2.2. □

Remark 3.4 (Scope of the Gaussian claim). The conclusion concerns the squared L^2 divergence between Gaussian-smoothed versions of the same base measure. The Gaussian kernel alone does not imply the Robinson property for an arbitrary divergence. A separate monotonicity argument would be required for each other divergence.

4 Implication for spectral scale selection

The eigenvector equation for a dissimilarity matrix reads

$$\rho v_i = \sum_{j=1}^{N} D_{ij} v_j.$$

It therefore rewards a scale for being far from scales that already carry high weight. Corollary 3.3 makes this precise for Gaussian KDEs and the squared L^2 divergence: the maximal Perron entry must occur at the smallest or largest bandwidth, regardless of whether an interior bandwidth is statistically preferable.

By contrast, a decreasing entrywise transform such as

$$S_{ij} = \exp\left(-\frac{D_{ij}}{\tau}\right), \qquad \tau > 0,$$

assigns large affinity to nearby scales and small affinity to distant scales. Its Perron vector measures weighted similarity rather than weighted dissimilarity. This is the precise reason to apply the similarity transformation before using Perron-based spectral scale selection.

Thesis-ready conclusion. For an ordered family lying on a common geodesic, geodesic convexity of the divergence forces the Perron maximum of the divergence matrix to an endpoint. For Gaussian KDEs with squared L^2 divergence, the bandwidth path is instead a strictly acute Hilbert-space chain; this stronger structure both makes the matrix a strict Robinson dissimilarity and yields the same endpoint conclusion. Thus using D itself creates a provable boundary bias in the setting relevant to Gaussian-kernel L^2 scale selection.

References

[1] M. Carmona, V. Chepoi, G. Naves, and P. Préa, Two simple but efficient algorithms to recognize Robinson dissimilarities, *Journal of Classification* 41 (2024), 455–479. doi:10.1007/s00357-023-09446-y.